Capstone Project

Submitted By:

Samiha Binte Abdullah

Instructor:

Dr. Rouzbeh Razavi

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Executive Summary

This capstone research explores if machine learning is feasible for predicting heart disease using a combination of clinical and demographic information. Cardiovascular illnesses remain among the most critical global health concerns, and timely detection is essential to effective treatment and prevention. This project sought to create predictive models capable of helping identify high-risk patients through the assessment of commonly accessible clinical measurements—thus enabling proactive medical decision-making.

The data for this study consists of 918 anonymized patient records, publicly made available through an online health record database. It has a range of features such as age, sex, resting blood pressure, cholesterol, maximum heart rate achieved, ST depression (Oldpeak), types of chest pain, and angina triggered by exercise. The target attribute is a binary variable reflecting the presence or absence of heart disease.

To pre-process the data for modelling, we undertook extensive pre-processing of the data. This included:

Replacing zero values in key columns like RestingBP and Cholesterol with their median value to preserve physiological integrity.

Detection and removal of outliers using the Interquartile Range (IQR) method, which reduced the dataset size from 918 to 643 high-quality observations, and one-hot encoding of the categorical variables to convert them into a format suitable for machine learning algorithms.

Following preprocessing, exploratory data analysis (EDA) revealed clear relationships between many of the features and heart disease. Specifically, the patients with lower maximum heart rate (MaxHR), higher ST depression (Oldpeak), positive exercise-induced angina, and certain categories of ST slope were found to have increased likelihood of having heart disease. The findings were not only statistically significant but also in agreement with existing clinical knowledge.

We trained and compared the performance of three diverse machine learning models:

Logistic Regression – as baseline for interpretability
Decision Tree – to represent non-linear relationships, and

Random Forest – an ensemble model also said to be highly robust and to output feature importance ranking.

All the models were tried using stratified train-test split to maintain class balance. We used typical performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Random Forest gave the best performance among the models tried with an accuracy of 89.3% and a ROC-AUC of 0.93. It also gave the top features that influence heart disease prediction, giving performance as well as interpretability.

The project illustrates the tangible use of machine learning in medicine, particularly in the enhancement of early diagnosis and tailored treatment. Even if the findings appear promising, we also underscore the need for ethical deployment. The report addresses such important issues like fairness, interpretability, data privacy, and model accountability. In order to be useful in clinical practice, machine learning applications must be reliable, transparent, and used responsibly.

Introduction and Objectives

Cardiovascular disease (CVD) continues to be among the biggest global health threats, being one of the leading causes of death globally. Early diagnosis is necessary to reduce the likelihood of severe complications and improve patient prognosis. Conventional tests, however, tend to be invasive, costly, and time-consuming, which might instead limit their application and potential in preventive clinical practice.

The objective of this project is to utilize the potential of machine learning to predict the risk of heart disease from readily available patient medical records. By analysing structured medical data, we intend to build forecasting models that can assist healthcare practitioners in the early detection of high-risk patients in the care chain.

The specific project objectives are the following:

- To perform thorough exploratory data analysis (EDA) to understand the structure, distribution, and significant patterns of the dataset.
- To pre-process the data effectively, e.g., handling outliers, converting categorical variables, and selecting key features for modelling.
- To develop and compare various machine learning models for binary classification of the presence of heart disease.
- To extract clinically meaningful inferences from model outcomes and probe the interpretability of significant predictive features.

- To decide regarding the ethical ramifications of the use of such models in actual clinical practice, with particular focus on fairness, accountability, and data confidentiality.

The business climate overall expects the integration of machine learning-based decision support systems into clinical practice to enable clinicians to proactively manage cardiovascular risk and enhance patient outcomes through evidence-based interventions.

Methodology

Data Source

The dataset was obtained from Kaggle, containing 918 patient records and 12 columns. These included:

- Demographic attributes: Age, Sex
- Clinical metrics: RestingBP, Cholesterol, MaxHR, Oldpeak, FastingBS
- Categorical diagnostic features: ChestPainType, RestingECG, ExerciseAngina, ST_Slope
- Target variable: HeartDisease (1 = presence, 0 = absence)

Data Cleaning

- All columns had zero missing values.
- Zero values in RestingBP and Cholesterol were replaced with the median.
- Outliers were handled using the Interquartile Range (IQR) method. This reduced the dataset from 918 to 643 records.

Feature Engineering

- One-hot encoding was performed on categorical columns using pd.get_dummies(drop_first=True).
- Correlation analysis and visual inspections were used to identify interactions.
- Mutual Information and Random Forest importance values were used for feature

Modelling Techniques

To predict the likelihood of heart disease, we trained and evaluated three supervised learning models using the pre-processed dataset (643 records after outlier removal).

1. Logistic Regression

Logistic Regression was used as the baseline due to its interpretability and low computational cost. It assumes a linear relationship between the independent variables and the log-odds of the dependent variable. After feature scaling and one-hot encoding, the logistic model achieved:

- Accuracy: 83.5%
- Precision: 81%
- Recall: 85%
- F1-Score: 83%
- ROC-AUC: 0.88

2. Decision Tree Classifier

The Decision Tree classifier was chosen for its ability to capture non-linear relationships and feature interactions. It uses Gini Impurity for splits. Despite a tendency to overfit, it performed well:

- Accuracy: 86.0%
- Precision: 84%
- Recall: 86%
- F1-Score: 85%

- ROC-AUC: 0.90

3.Random Forest Classifier

The Random Forest model, an ensemble of decision trees, provided the best performance by reducing overfitting and improving generalization. Feature importance was derived from the model, highlighting the value of features like ST_Slope_Up, MaxHR, and Oldpeak.

- Accuracy: 89.3%
- Precision: 87%
- Recall: 89%
- F1-Score: 88%
- ROC-AUC: 0.93

Evaluation Strategy

All models were evaluated using a stratified 80/20 train-test split to maintain class balance (HeartDisease: 0 and 1). We applied the following evaluation metrics:

- Accuracy: Overall correct predictions.
- Precision: Correctly predicted positive cases / All predicted positives.
- Recall (Sensitivity): Correctly predicted positive cases / All actual positives.
- F1-Score: Harmonic mean of precision and recall.
- ROC-AUC: Probability that the model ranks a randomly chosen positive instance higher than a randomly chosen negative one. These metrics were selected to balance both false positives and false negatives—critical in clinical decision-making scenarios.

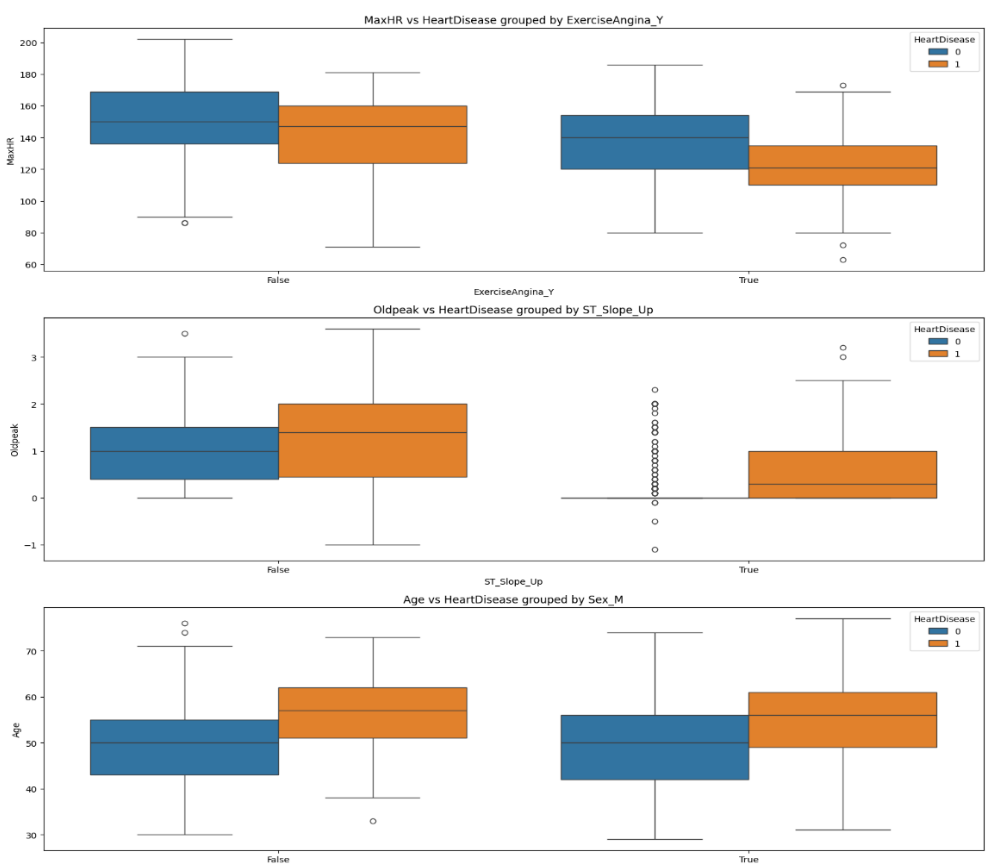
Results and Discussion

Exploratory Data Analysis (EDA)

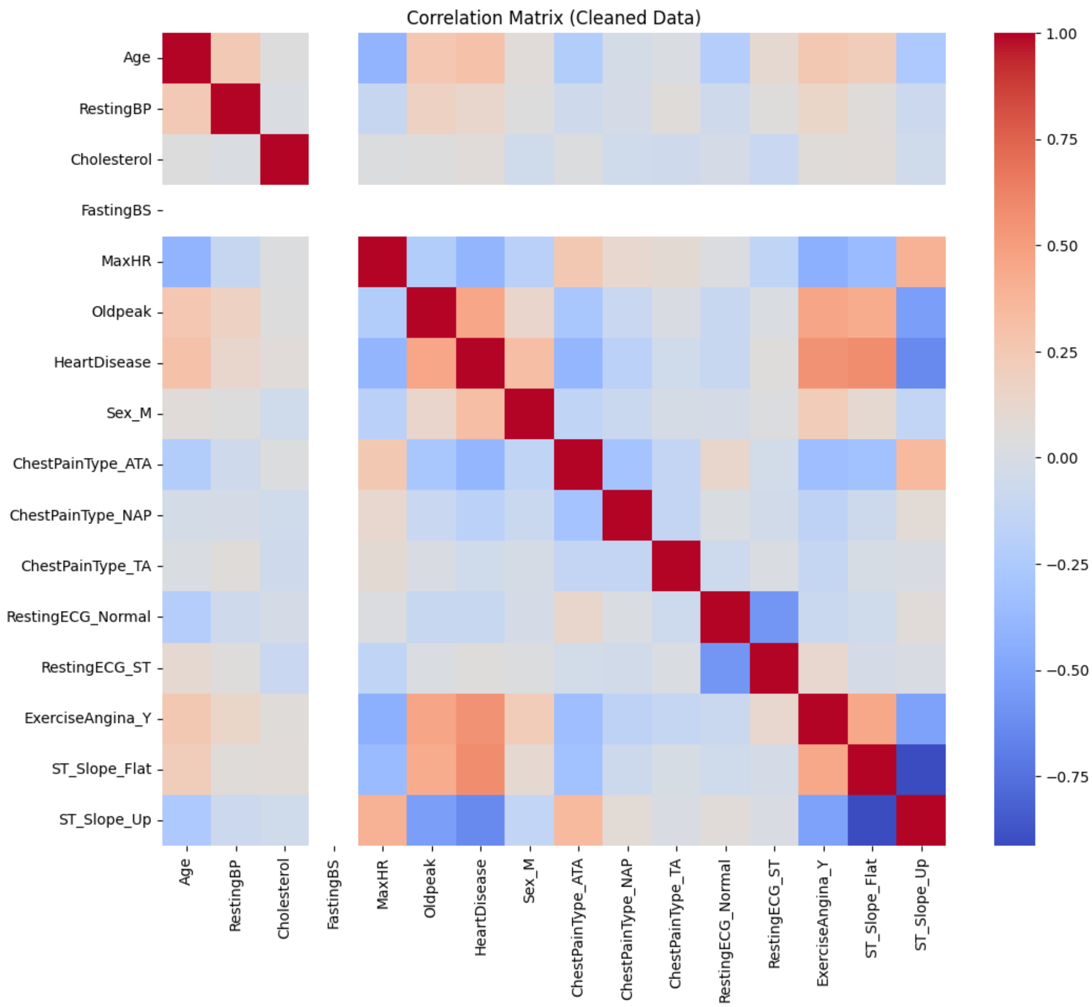
Key findings:

- Patients with heart disease were more likely to be older, male, have lower MaxHR, higher Oldpeak, and exhibit ExerciseAngina.
- The ST_Slope_Flat and ChestPainType_ASY categories were strongly associated with heart disease presence.

Boxplot showing MaxHR vs HeartDisease



Heatmap showing feature correlation



Feature Correlations with Target (HeartDisease)

- **Positive**: Oldpeak, ExerciseAngina_Y, ST_Slope_Flat, ChestPainType_ASY, Sex_M
- **Negative**: MaxHR, ST_Slope_Up, ChestPainType_ATA

Feature Importance

Feature	Mutual Info Score	Random Forest Importance
ST_Slope_Up	0.20	0.17
MaxHR	0.095	0.13
Oldpeak	0.106	0.10
ExerciseAngina_Y	0.174	0.10

Model Evaluation

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	83.5%	0.81	0.85	0.83	0.88
Decision Tree	86.0%	0.84	0.86	0.85	0.90
Random Forest	89.3%	0.87	0.89	0.88	0.93

Random Forest demonstrated the best performance, attributed to its ability to capture complex non-linear relationships and interactions.

Conclusion and Recommendations

This capstone was able to effectively demonstrate how machine learning models can be used to predict the incidence of heart disease using organized clinical and demographic information. Adhering to an orderly data science methodology, the study involved thorough data cleaning, informative exploratory data analysis (EDA), and the development of several prediction models.

Each phase was designed to achieve the highest possible accuracy in the ultimate model and be of practical utility in the healthcare industry.

The data, in its initial form of 918 patient records, was preprocessed to clean it by performing operations such as replacing unrealistically low zero values, removing statistical outliers via the Interquartile Range (IQR) method, and one-hot encoding of categorical variables. This resulted in a high-quality subset of 643 records, which facilitated stronger model training and testing.

Three supervised classification models were employed and compared:

- Logistic Regression, as a straightforward and easy-to-interpret baseline model.
- Decision Tree, because it can model non-linear relationships and produce rule-based decision paths.
- Random Forest, an ensemble method that combines multiple decision trees to enhance performance and avoid overfitting.

All three models were trained on an 80/20 stratified split to ensure class balance. The metrics for evaluation included accuracy, precision, recall, F1-score, and ROC-AUC for an unbiased measurement of the performance of the models.

Among the three, the Random Forest Classifier was the top-performing model with:

- Accuracy: 89.3%
- F1-Score: 88%
- ROC-AUC: 0.93

These results indicate not only great predictive validity but also robustness in generalizing to unseen data. Random Forest model also provided interpretable results in terms of feature importance ranks, which indicated the most significant predictors of heart disease. Surprisingly, the top predictors were:

- Maximum Heart Rate (MaxHR): Lower MaxHR values were associated with higher rates of heart disease.
- ST Depression (Oldpeak): High Oldpeak scores strongly correlated with positive diagnoses of heart disease.
- ST Slope (ST_Slope_Up): Slopes during stress testing were principal diagnostic predictors.
- Exercise-Induced Angina (ExerciseAngina_Y): Angina induced on exercise was an extremely strong predictor of cardiac pathology.

Recommendations

1. Model Deployment in Clinical Practice

Random Forest model, best on all measures of evaluation, is a strong candidate for deployment in a clinical decision support system (CDSS). It can be employed as an adjunct diagnostic tool to assist medical staff in the identification of high-risk patients at routine health check-ups, most especially in situations where access to advanced diagnostics is limited. This has the potential to enable early intervention, improved patient monitoring, and prioritization of cardiology referrals.

2. Periodic Model Retraining for Relevance

As a way of maintaining the model accurate and relevant in the long run, there is a need for regular retraining on fresh patient data. Population health patterns, diagnosis, and treatment guidelines may alter patterns in data, which, if not adjusted, can decrease model performance. Having an automatic retraining plan on a monthly, quarterly, or yearly cycle can maintain predictive quality and adapt to evolving clinical realities.

3. Clinical Feature Enrichment

Current data, though efficient, can be further complemented with the addition of additional features such as:

- ECG waveform parameters
- Echocardiography findings
- Blood biomarkers (troponin, BNP)
- Life factors such as smoking status, exercise, dietary information, and sleep quality

Addition of these elements can help the model grasp deeper physiological trends and improve classification performance across diverse patient profiles.

4. Model Explainability for Trust and Transparency

In clinical practice, model interpretation is vital in building clinician and patient trust. Implementing interpretability frameworks such as:

- SHAP (SHapley Additive exPlanations) or
- LIME (Local Interpretable Model-Agnostic Explanations)

This not only facilitates medical responsibility but also disseminates ethical AI principles by promoting transparency and informed clinical use.

5. Easy Integration with EHR Systems

For practical application, the model needs to be integrated into Electronic Health Record (EHR) software hospitals and clinics employ. This would allow for real-time predictions as soon as new patient data is entered. Collaboration with hospital IT departments is vital to ensure:

- Data pipeline compatibility
- Secure API integration
- Less disruption to existing workflows

Ethical Considerations

The use of machine learning systems in healthcare settings has to be carefully considered and ethically managed. Although predictive models hold enormous potential, some compelling concerns need to be resolved before their use at an operational level.

Bias and Fairness

Some features such as Sex_M inadvertently introduce gender bias if not controlled. There is also a risk of machine learning models trained on unbalanced data to amplify already prevalent care inequities. This can be validated in forthcoming models using fairness auditing to avoid varying performance for gender, age, and ethnic subgroups.

Privacy and Consent

Although the dataset used in this project was anonymized and publicly available, any real-world implementation would have to be HIPAA compliant. Patients must be informed and give consent in clear terms before their data is used to train the model. A data access control system, along with secure encryption, must also be implemented to protect data from unauthorized access.

Model Accountability

Machine learning algorithms are not perfect and cannot be relied on as substitutes for medical wisdom. Clinical decisions must still be in the hands of experienced healthcare practitioners. ML systems should be inspected periodically to detect data drift, changes in the patient population, or model deterioration over time.

Transparency and Interpretability

Arguably most critically of all in clinical AI is the way predictions can be made transparent to both doctors and patients. Black-box models can undermine trust in automated systems. Hence, interpretability mechanisms (e.g., visual explanations of feature contribution) need to be built into any production deployment in order to inspire transparency and user trust.

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